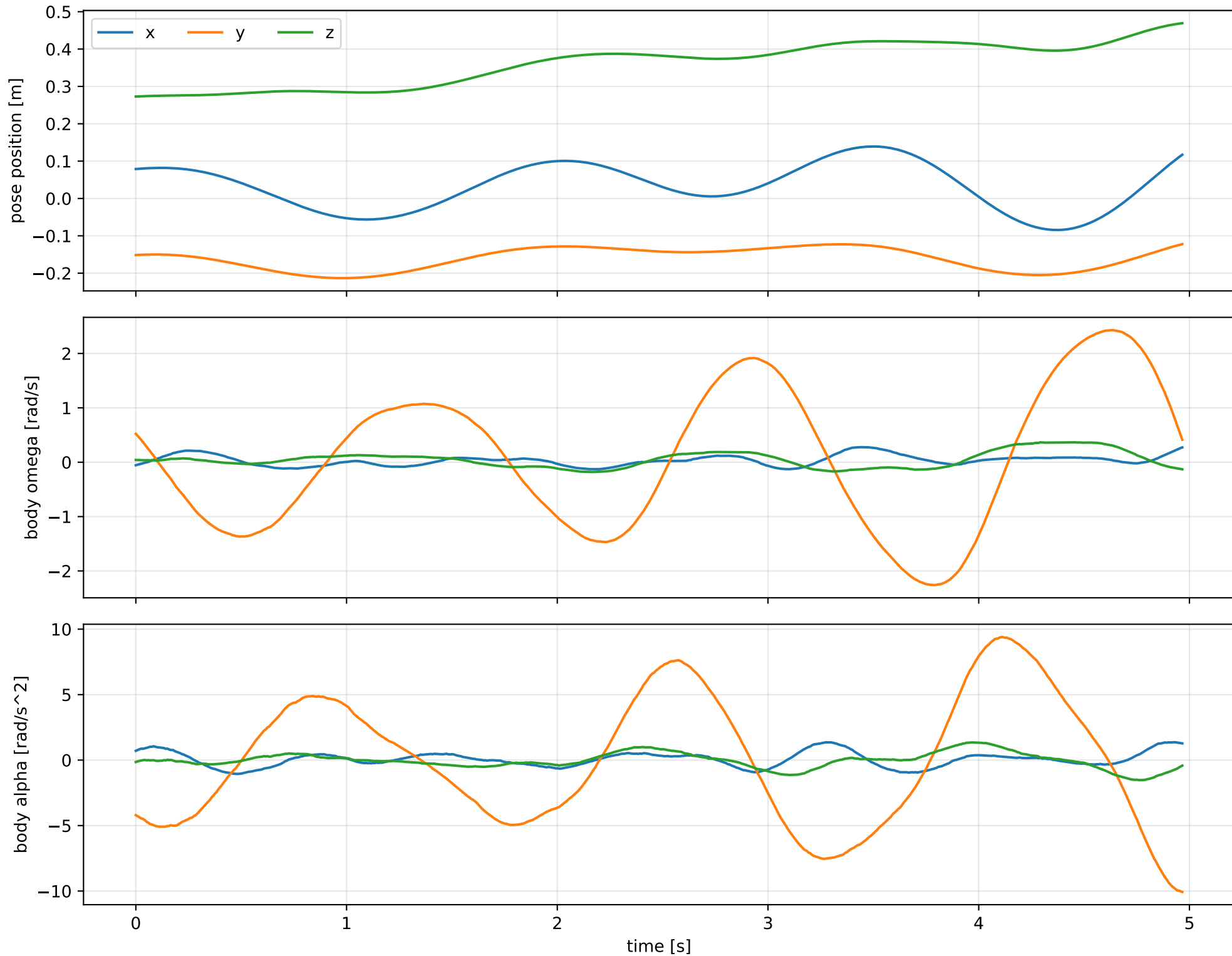
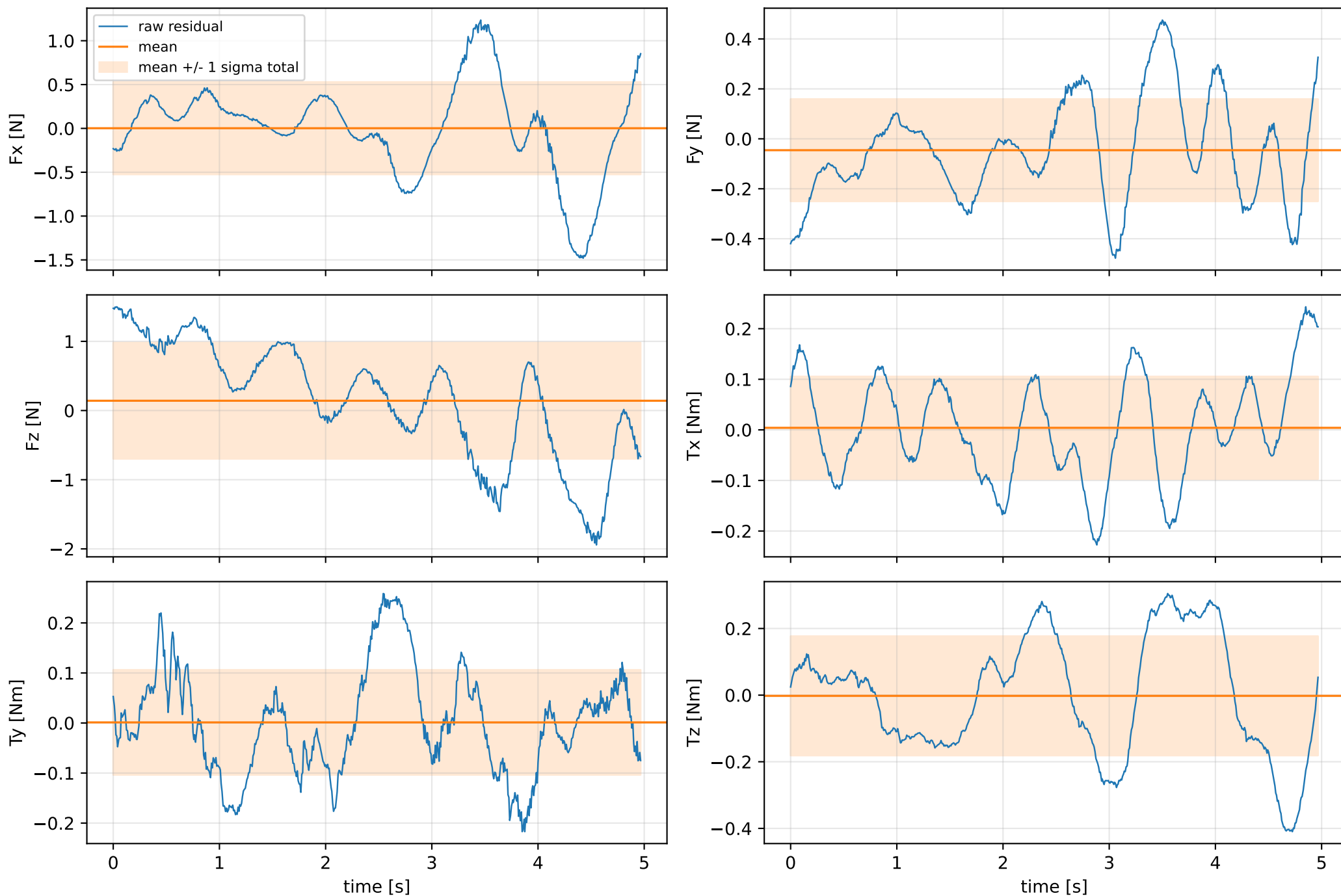


inertia_over_mass_yz_nominal: SG trajectory and derived motion



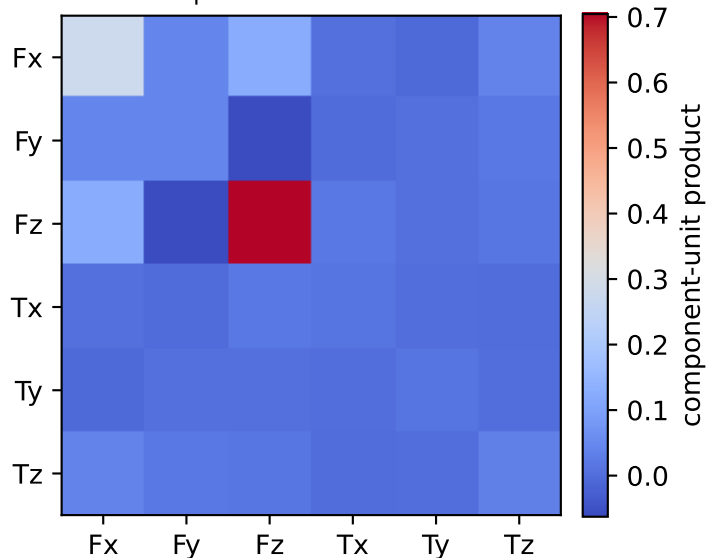
inertia_over_mass_yz_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



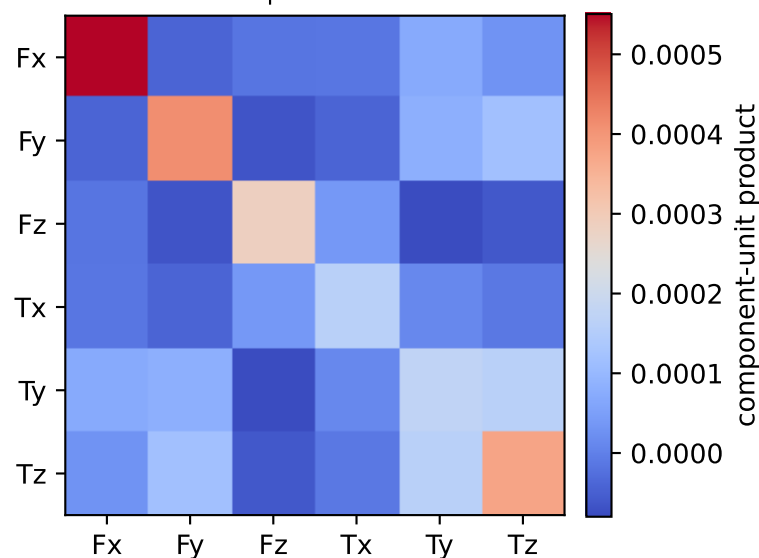
vector RMS about zero: force 1.0219 N, torque 0.23079 Nm; component std about mean: force [0.5268 0.2047 0.8393], torque [0.1014 0.1049 0.179]

inertia_over_mass_yz_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.009 0.016 0.035 0.266 0.741]

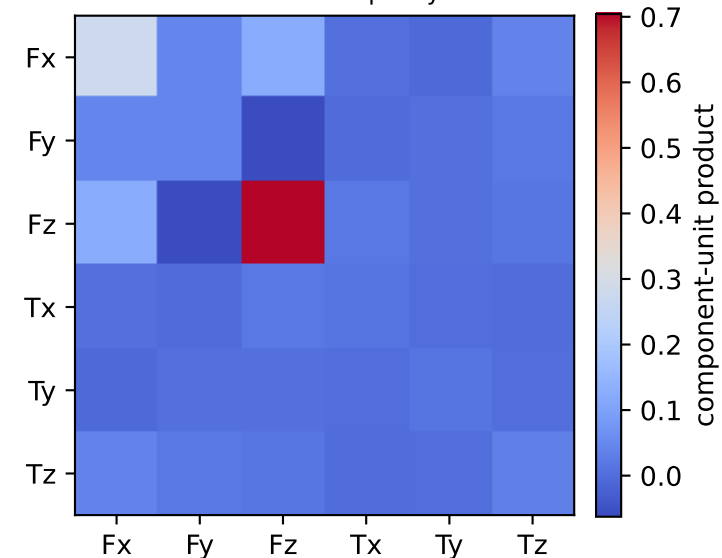
empirical total covariance



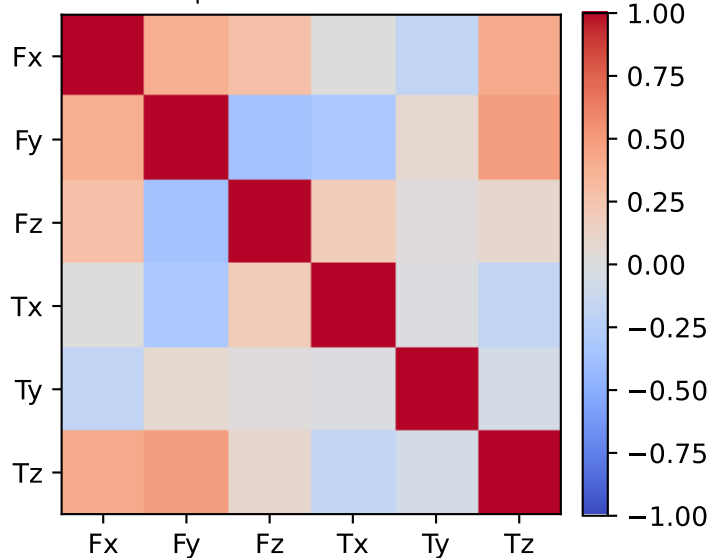
mean full-SG prediction covariance



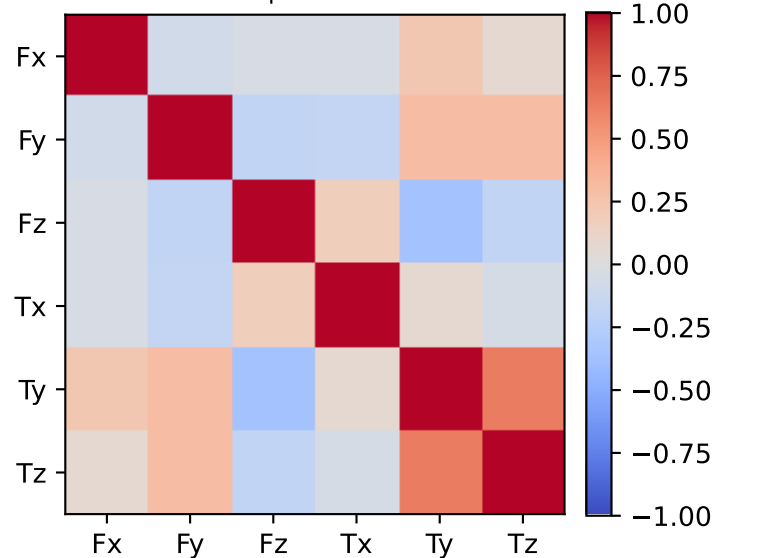
PSD excess model discrepancy covariance



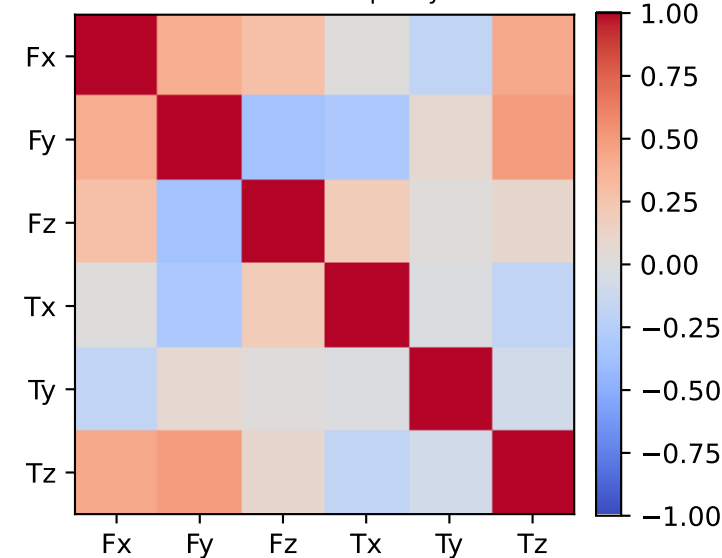
empirical total correlation



mean full-SG prediction correlation

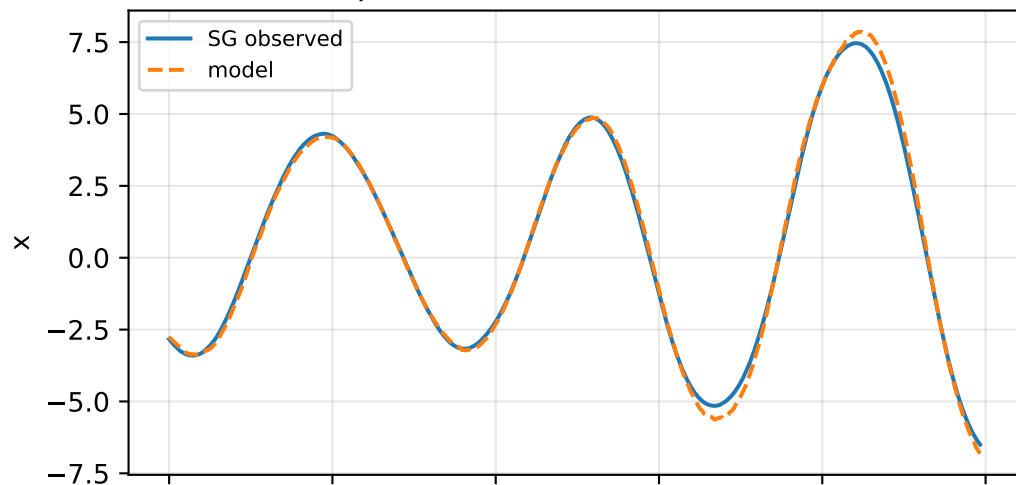


PSD excess model discrepancy correlation

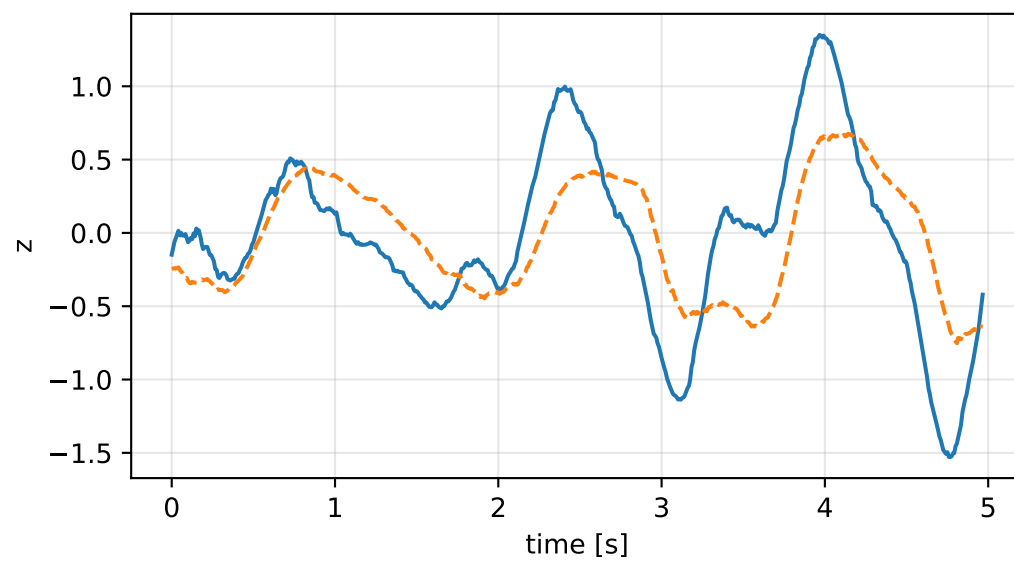
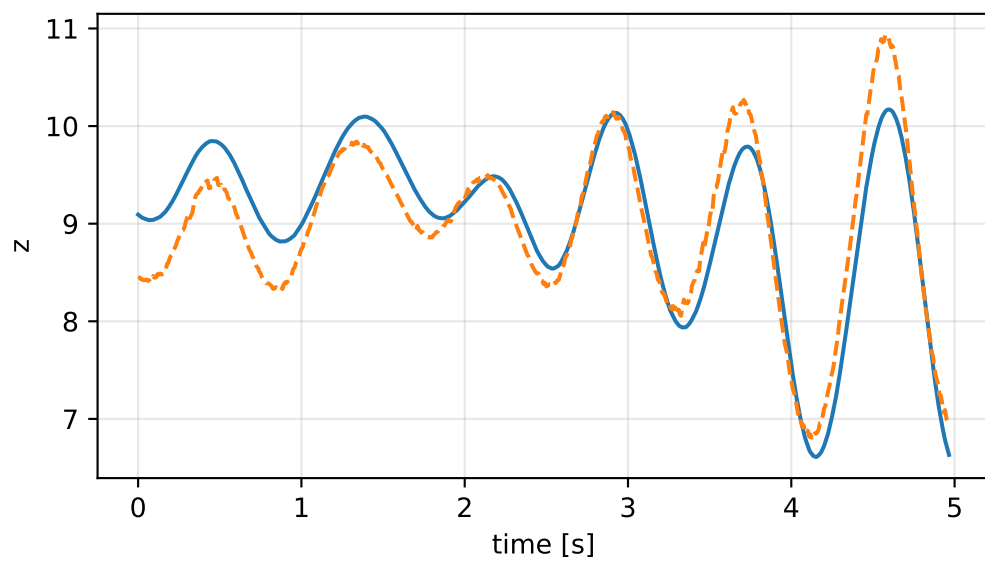
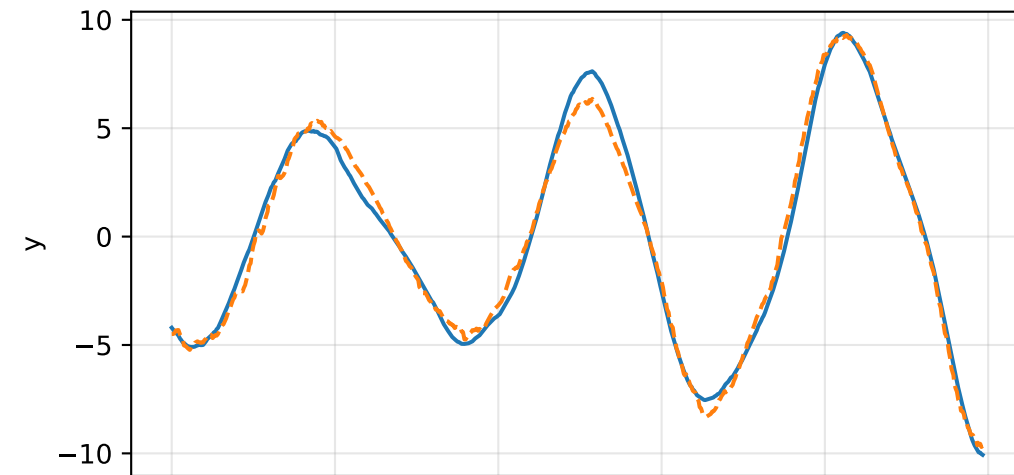
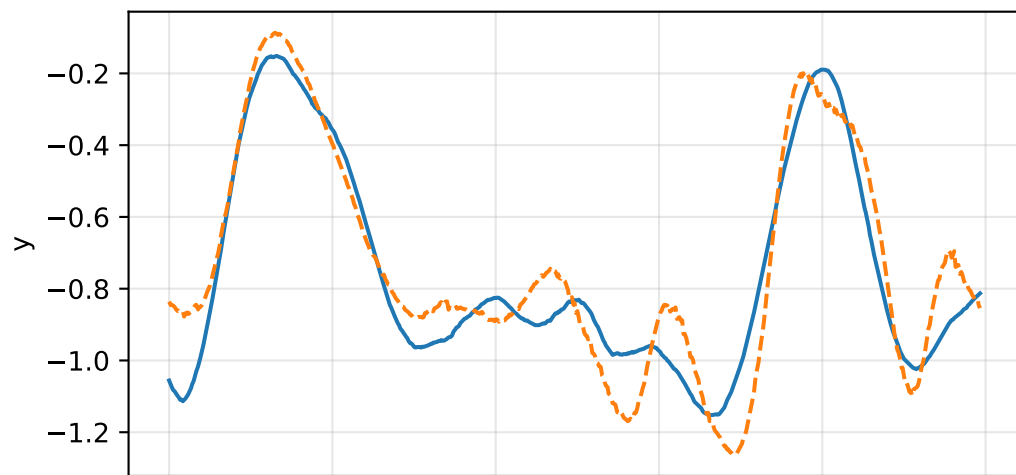
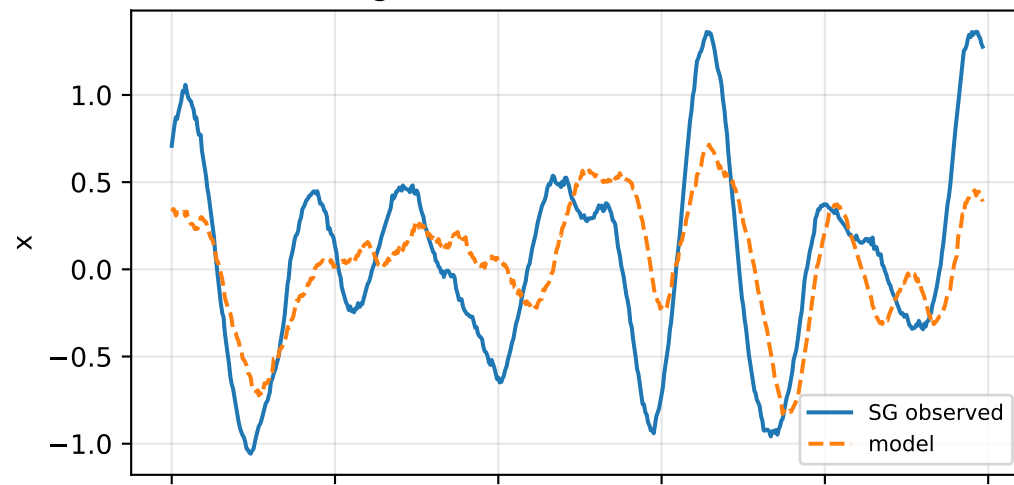


inertia_over_mass_yz_nominal: acceleration residual objective

specific acceleration [m/s^2]

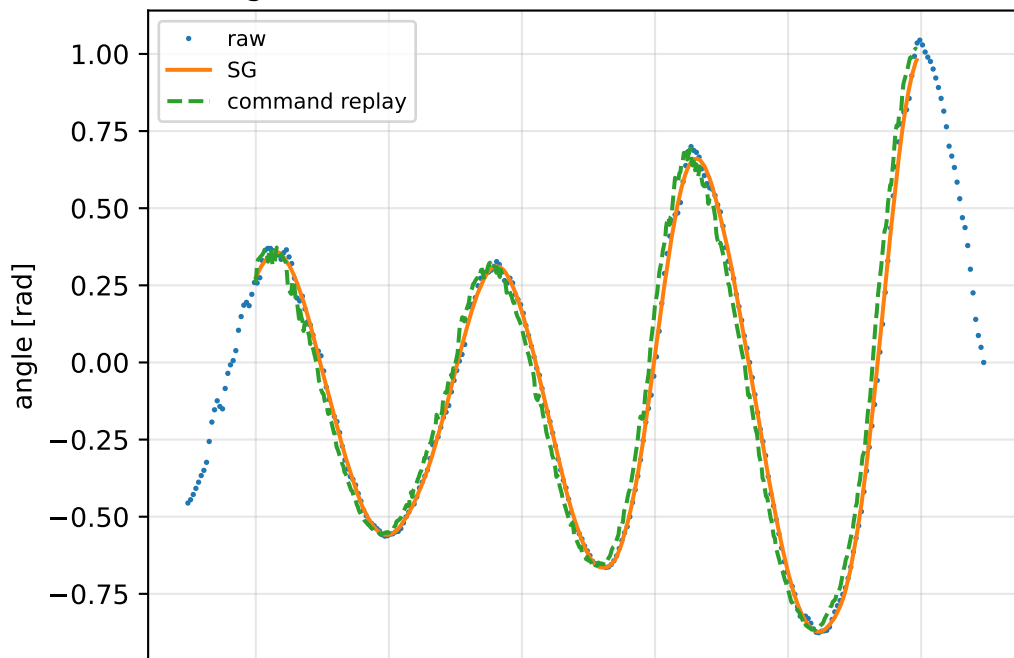


angular acceleration [rad/s^2]

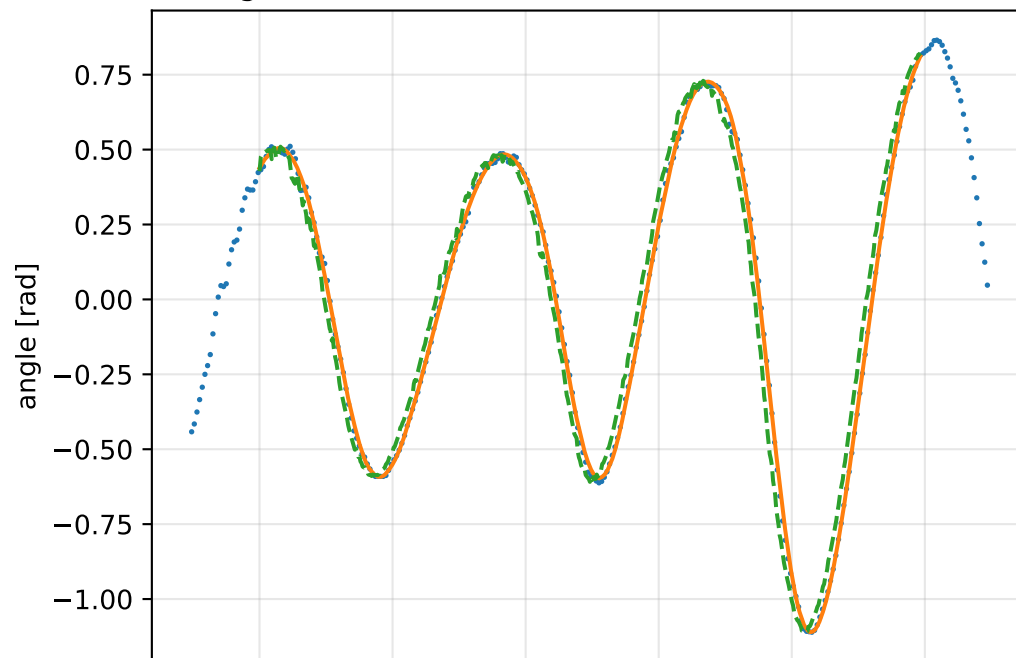


inertia_over_mass_yz_nominal: gimbal measurement audit (objective=measured_sg)

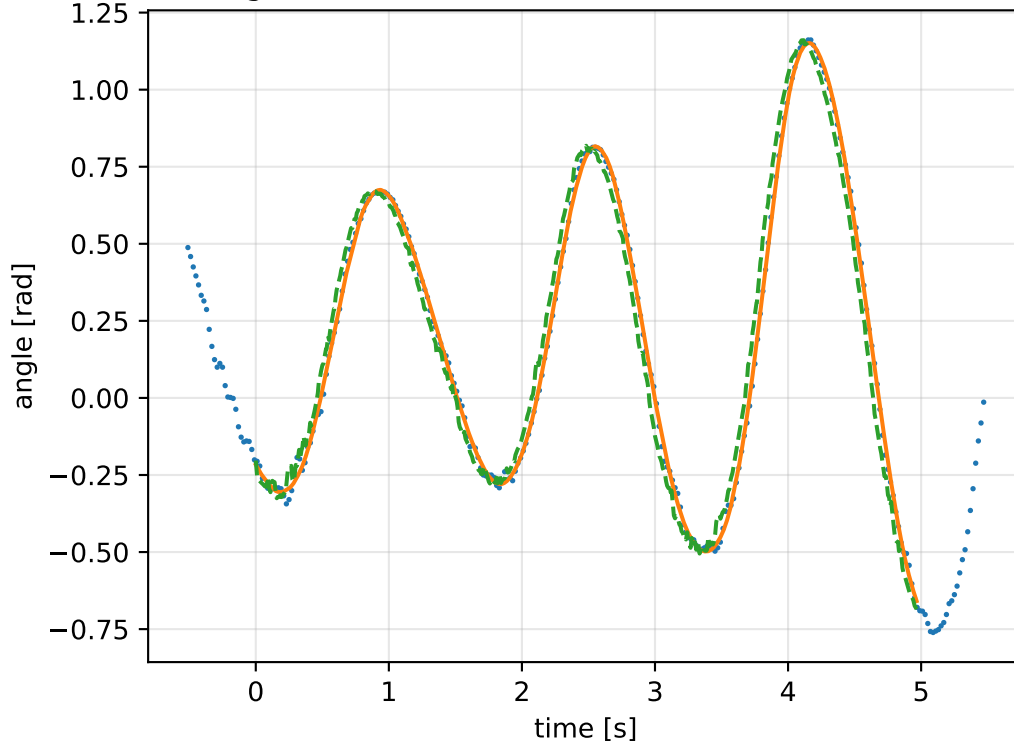
gimbal 1: RMSE=0.0759, max=0.2095 rad



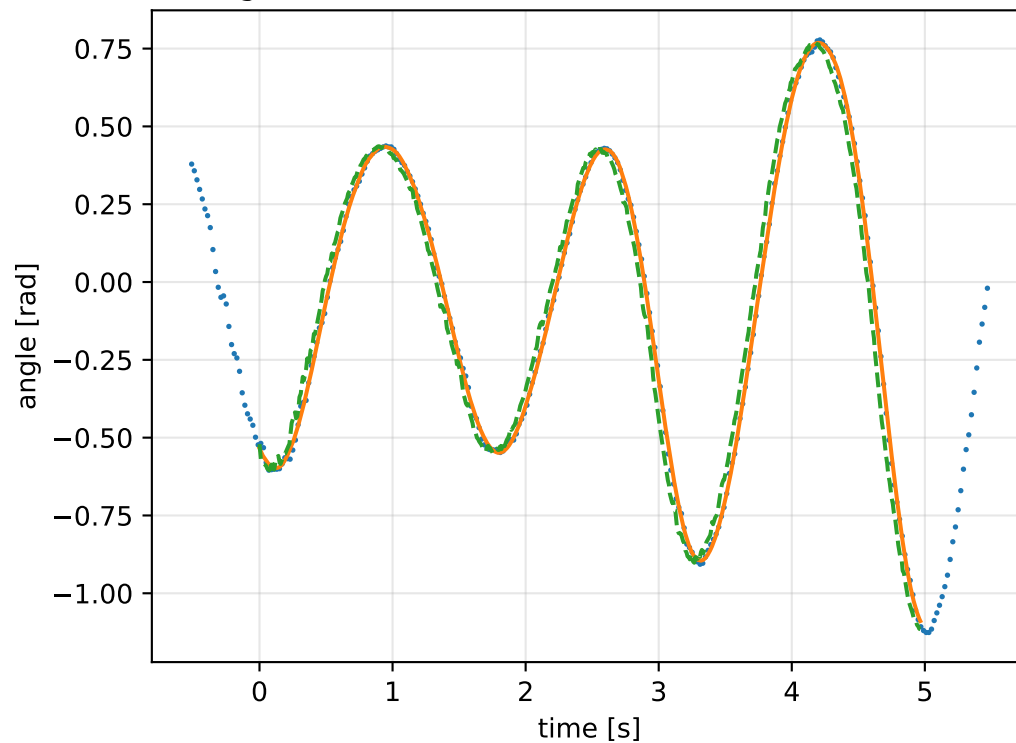
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

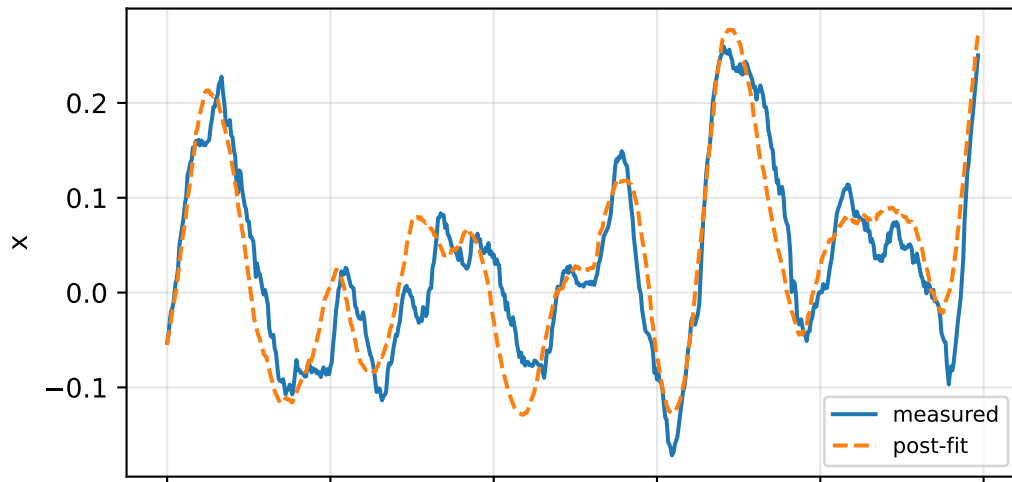


gimbal 4: RMSE=0.07131, max=0.1737 rad

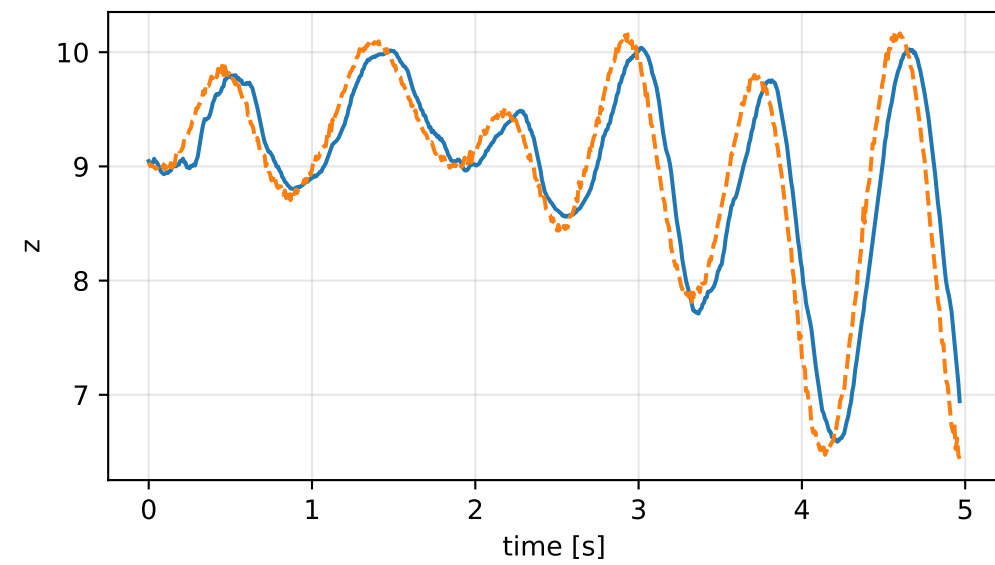
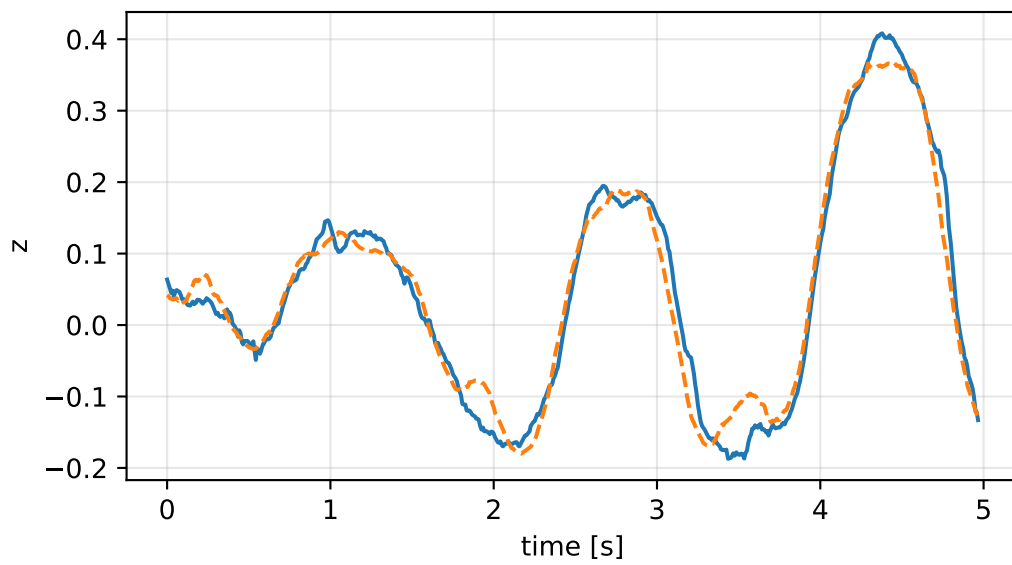
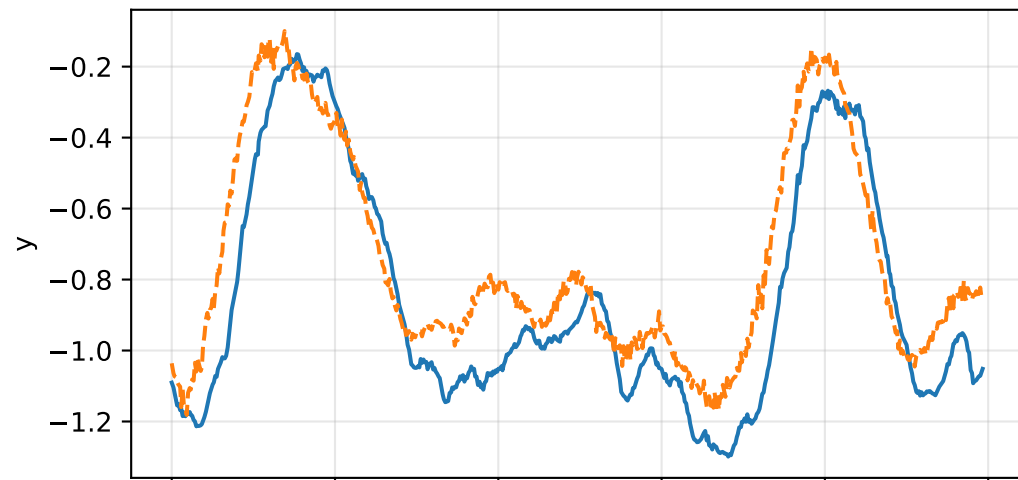
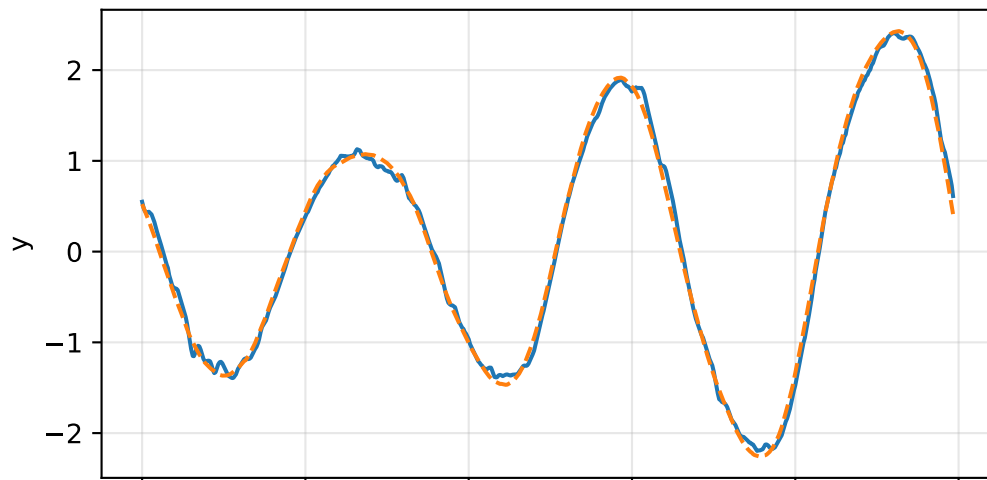
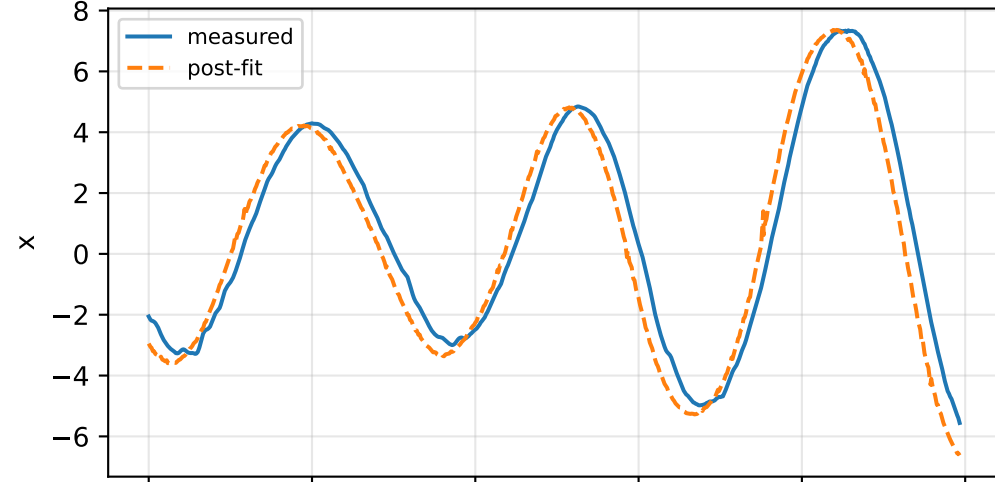


inertia_over_mass_yz_nominal: IMU comparison (diagnostic only)

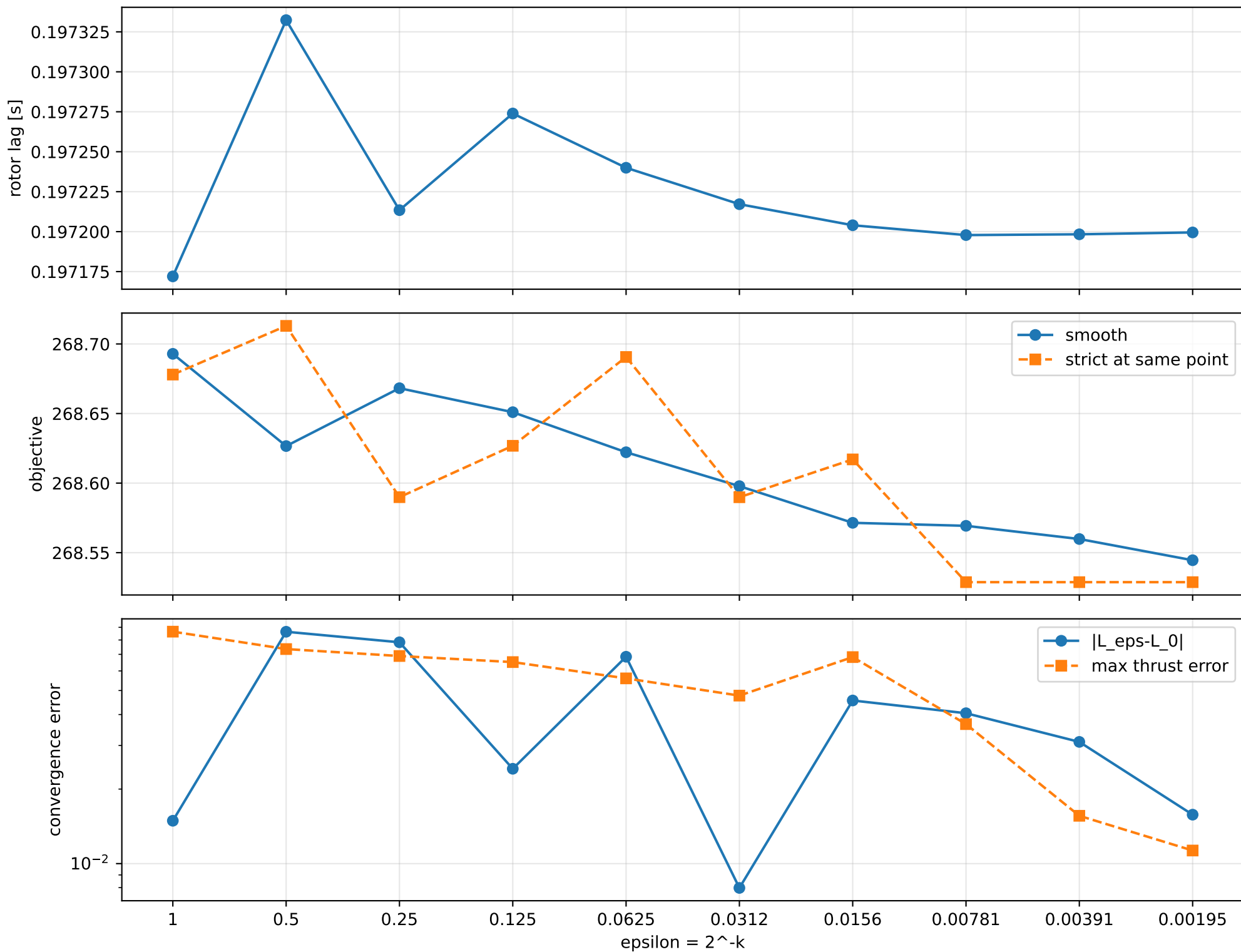
gyro [rad/s]



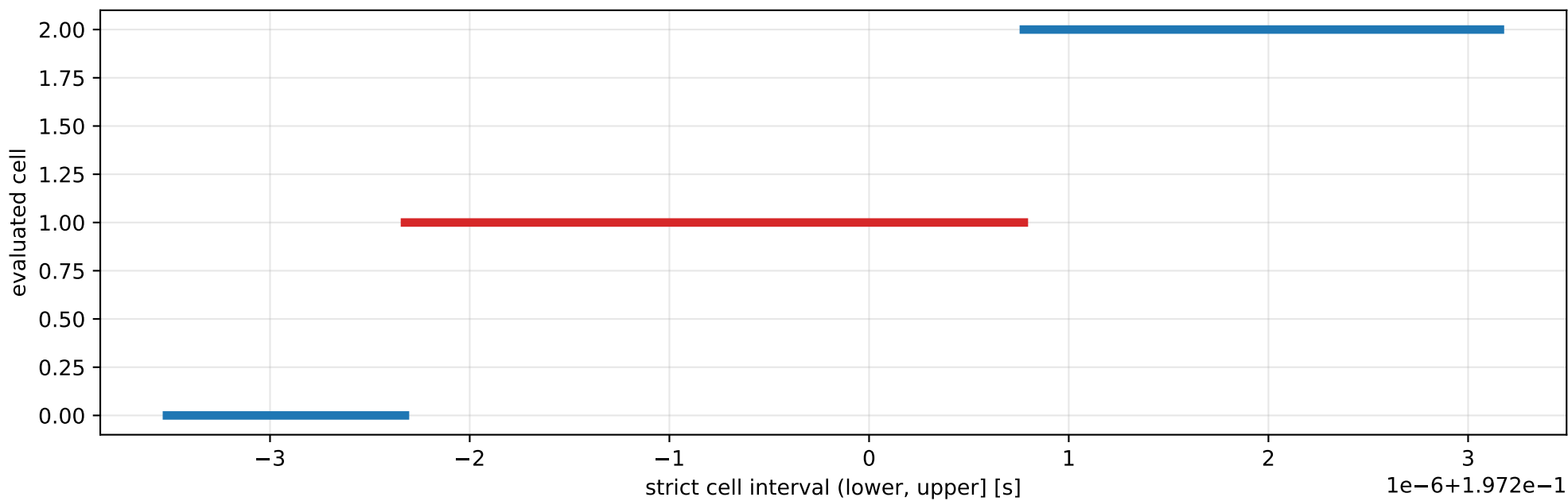
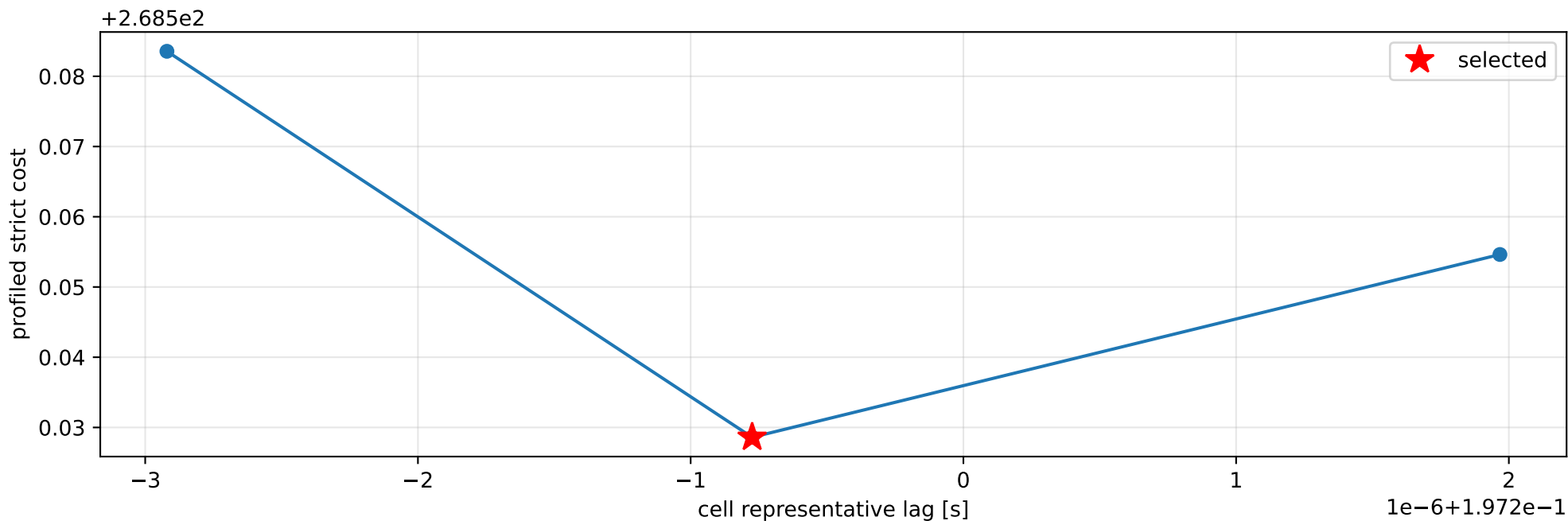
specific force [m/s^2]



inertia_over_mass_yz_nominal: smooth-to-strict continuation



inertia_over_mass_yz_nominal: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199511 s; data boundary=False

inertia_over_mass_yz_nominal: parameter estimate

Case: inertia_over_mass_yz_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.80919852

inertia [kg m²]:

[[3.86961865e-01 -4.56163345e-04 -6.28943920e-02]

[-4.56163345e-04 2.94670531e-01 9.58313158e-08]

[-6.28943920e-02 9.58313158e-08 6.53427945e-01]]

CoG body [m]: [0.01761515 -0.00409941 -0.04437641]

force effectiveness: [1.27571096 1.64258787 1.43454944 1.1895568]

scale-free J/m [m²]:

[[1.01586164e-01 -1.19753104e-04 -1.65111878e-02]

[-1.19753104e-04 7.73576197e-02 2.51578686e-08]

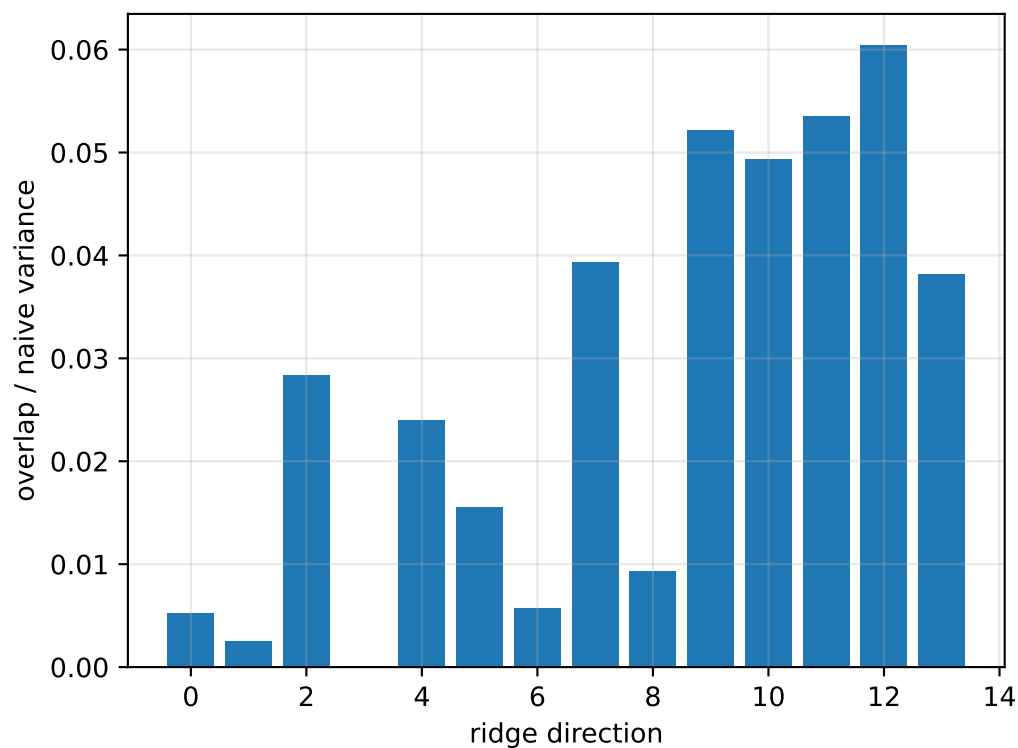
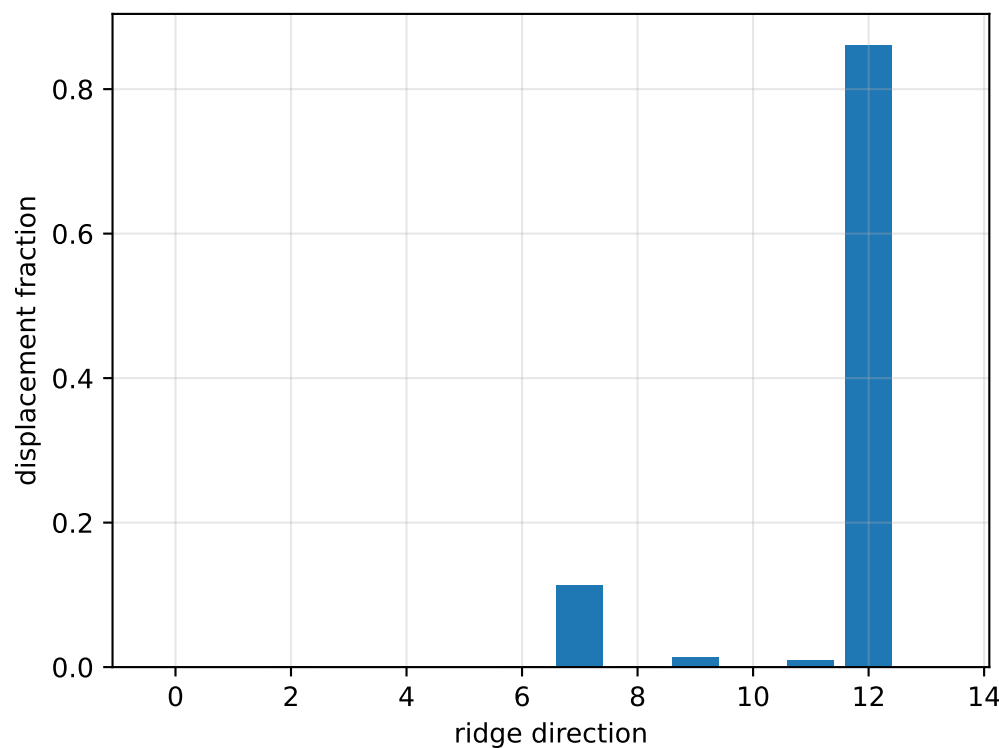
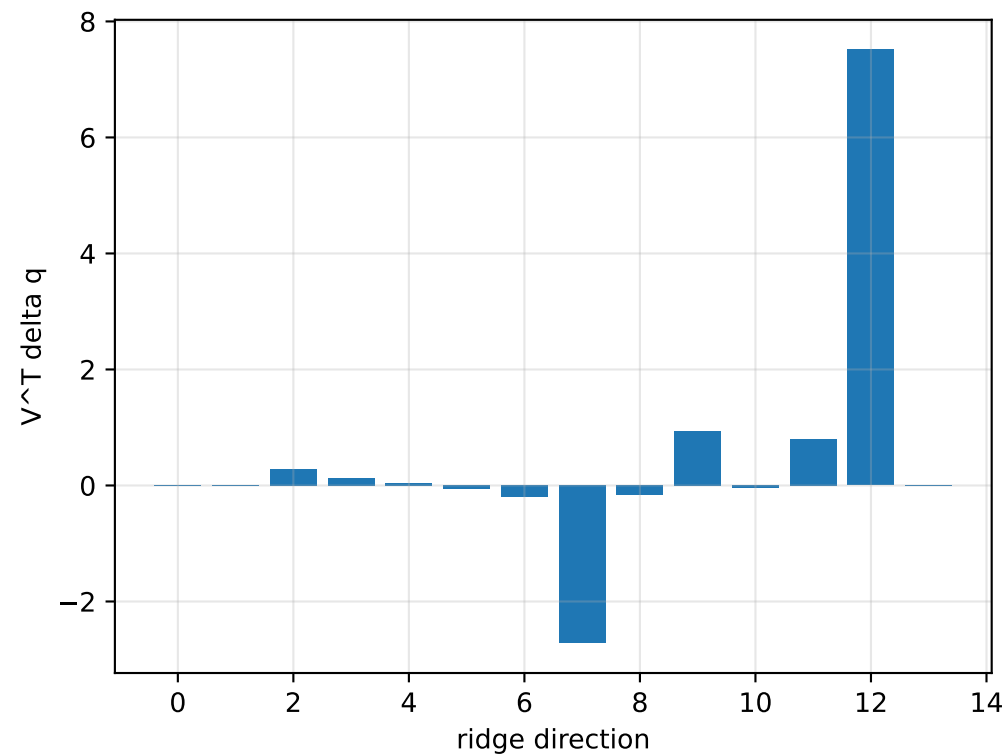
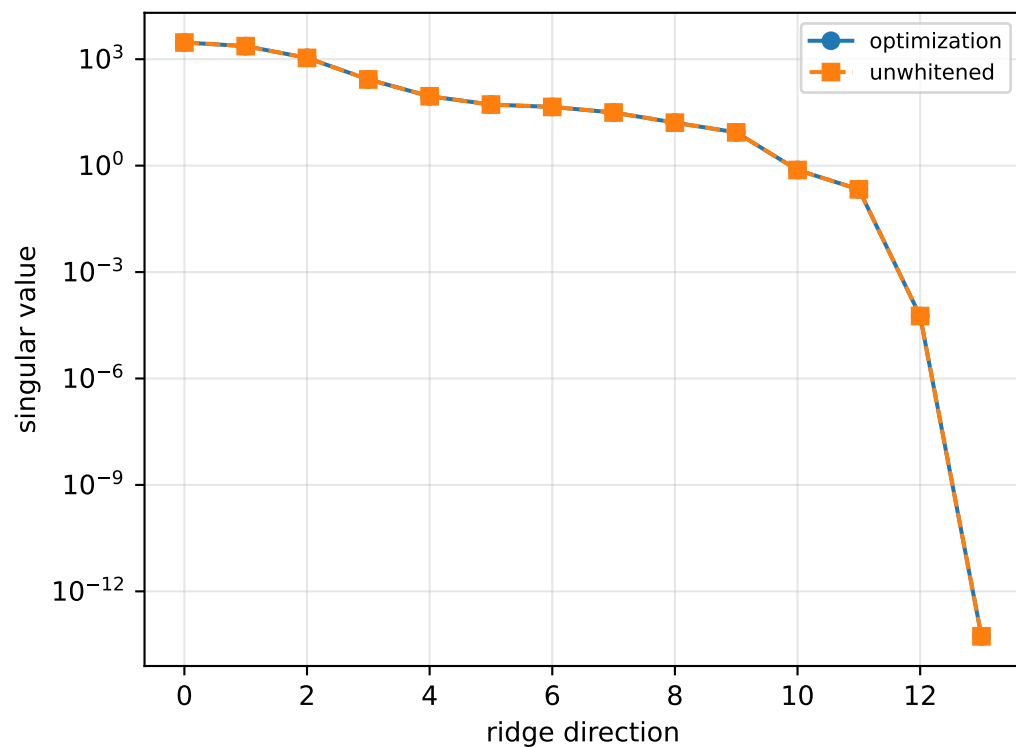
[-1.65111878e-02 2.51578686e-08 1.71539483e-01]]

scale-free f/m: [0.33490272 0.43121614 0.37660139 0.31228533]

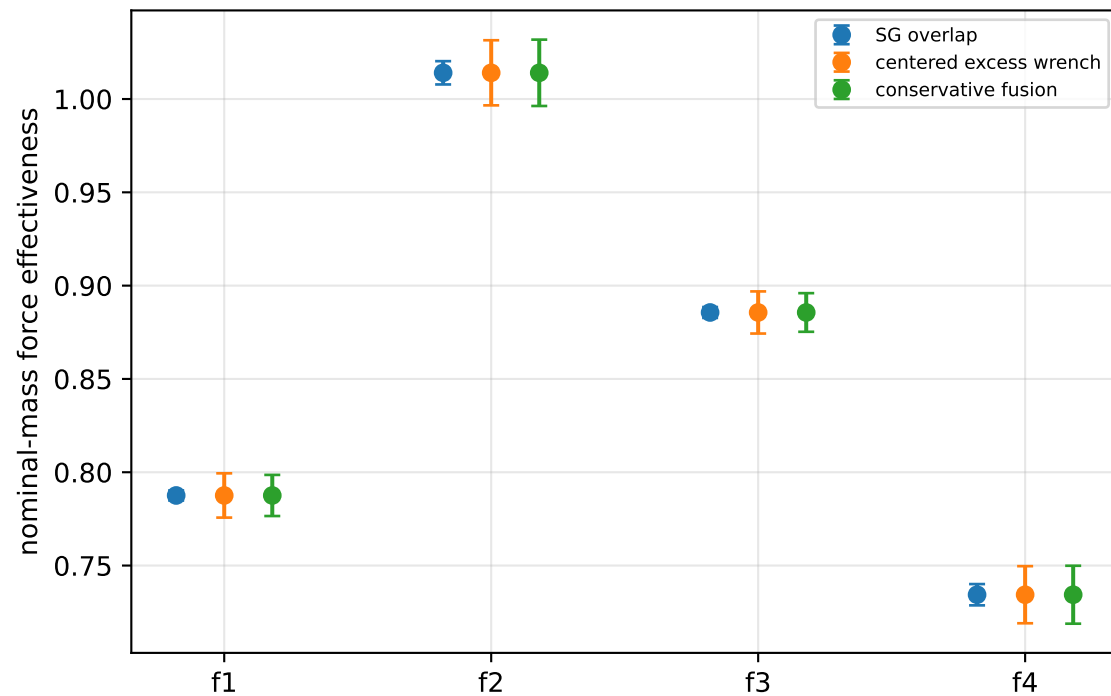
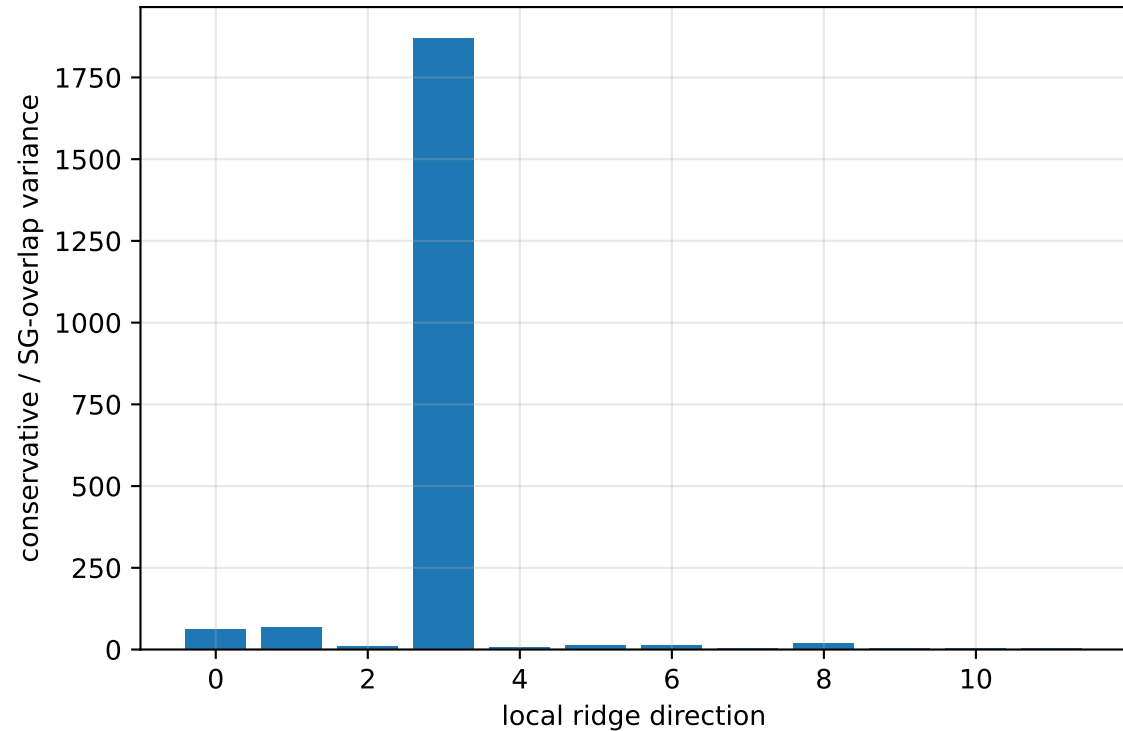
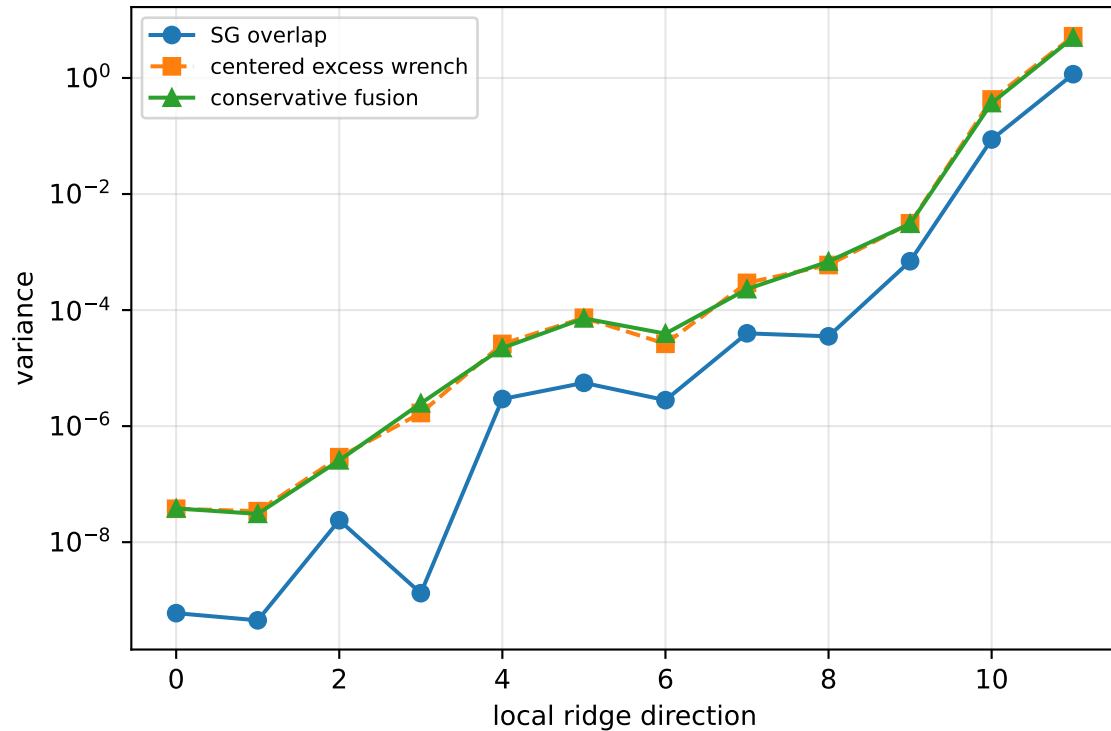
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

inertia_over_mass_yz_nominal: ridge spectrum (rank 13, nullity 1, ||v||=1.22e-13)



inertia_over_mass_yz_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 44.2$
wrench-to-acceleration recovery max error = $3.86\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

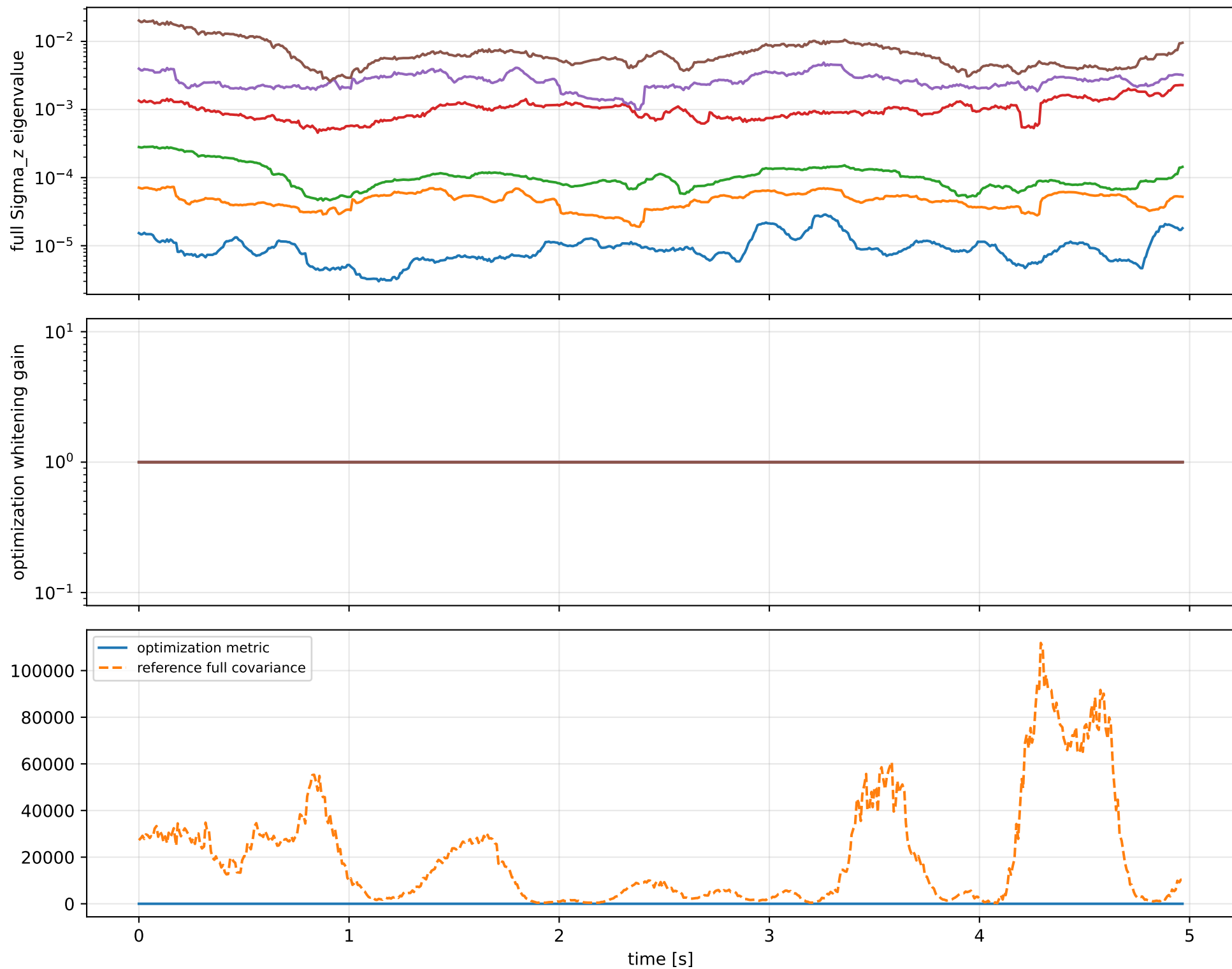
index 11: variance=4.967, ratio=4.273
 $\log_{\text{mass}} = -0.045$; second_moment_diag_1=-0.419
second_moment_diag_2=-0.0435; second_moment_diag_3=+0.687
second_moment_offdiag_12=+0.0463; second_moment_offdiag_13=-0.581
second_moment_offdiag_23=-0.0274; cog_x=+6.53e-06
cog_y=-0.000185; cog_z=+1.96e-05
log_force_eff_1=-0.0446; log_force_eff_2=-0.0441
log_force_eff_3=-0.046; log_force_eff_4=-0.0455

index 10: variance=0.3695, ratio=4.244
 $\log_{\text{mass}} = +0.00241$; second_moment_diag_1=+0.0223
second_moment_diag_2=+0.0104; second_moment_diag_3=-0.0408
second_moment_offdiag_12=+0.889; second_moment_offdiag_13=+0.0266
second_moment_offdiag_23=-0.454; cog_x=+0.0013
cog_y=-0.00325; cog_z=+0.000311
log_force_eff_1=+0.0098; log_force_eff_2=+0.0188
log_force_eff_3=-0.00459; log_force_eff_4=-0.0183

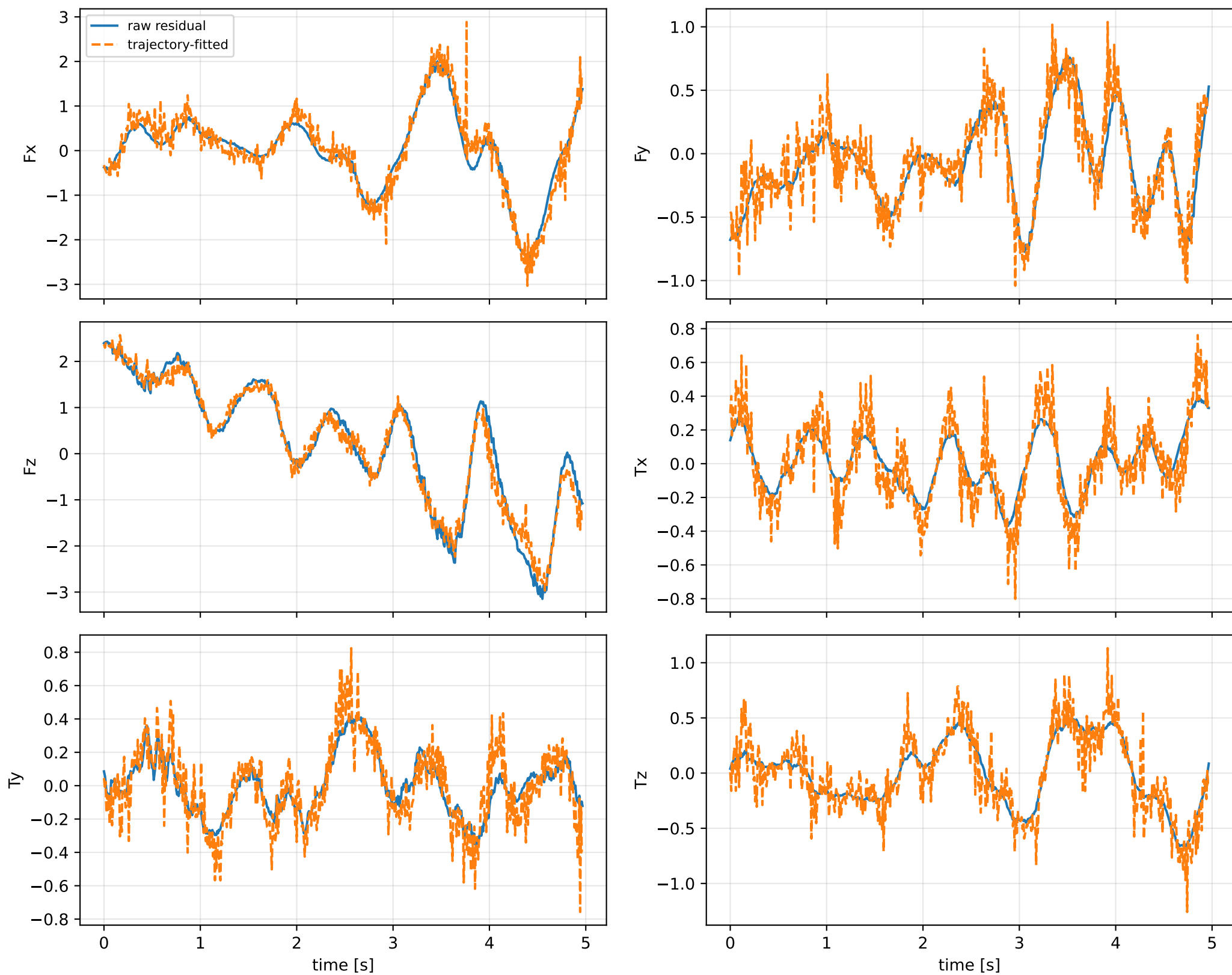
index 9: variance=0.003071, ratio=4.424
 $\log_{\text{mass}} = -0.148$; second_moment_diag_1=-0.033
second_moment_diag_2=+0.899; second_moment_diag_3=-0.0841
second_moment_offdiag_12=-0.00872; second_moment_offdiag_13=-0.0835
second_moment_offdiag_23=+0.0134; cog_x=+0.0221
cog_y=-0.0136; cog_z=-0.00393
log_force_eff_1=-0.187; log_force_eff_2=-0.022
log_force_eff_3=-0.0938; log_force_eff_4=-0.331

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_yz_nominal: optimization vs reference covariance



inertia_over_mass_yz_nominal: wrench / closure (force max 1.07e-14, torque max 1.14e-15)



inertia_over_mass_yz_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_yz_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: cb359687f62672f99943f224e046f1d453f420c78c59be95963f7c1e6ba83846

data objective: 268.528602458

prior objective: 3.38854791983e-12

total objective: 268.528602458

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_yz_nominal (inertia_over_mass_m2)

components: yz

target: [2.51604719e-08]

std: [1.e-06]

final: [2.51578686e-08]

error: [-2.60328559e-12]

standardized residual: [-2.60328559e-06]

factor objective: 3.38854791983e-12